

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
M.Tech.	Computer Science & Engineering	IIT Gandhinagar	2025-Present	8.4
B.Tech.	Computer Engineering	FCRIT	2021-2025	9.38
Class XII	Physics, Chemistry, Maths	Kendriya Vidyalaya	2020-2021	88.4
Class X		Kendriya Vidyalaya	2018-2019	89.4

INTERNSHIPS

AI/ML Developer Intern, TranquilMind, Avisk Pvt. Ltd.

[August '23 - April '24]

Built a patient-centric mental healthcare system using NLP and CBT-based dialogue modeling, improving user experience by 15%.

Developed an end-to-end AI/ML pipeline with a Node.js + MongoDB backend for secure, privacy-preserving healthcare deployment.

Designed interfaces in React Native and Power Apps, increasing average user session duration and engagement.

M.TECH THESIS - Conversational System for Indic Languages

Graduate Research Student, Lingo Research Group, IIT Gandhinagar

[Jan '26 - Present]

(Advisor - Prof. Mayank Singh, IIT Gandhinagar) | [Paper](#) | [Hugging Face](#)

Built IndicTalk, India's first and largest multilingual, code-mixed, persona-based conversational dataset of size 1,328,602 conversations, spanning 9 Indic languages in native & Roman scripts using LLM self-play with reflection, validation, and dynamic multi-turn generation.

Designing & evaluating an end-to-end multilingual conversational system by fine-tuning the model on the IndicTalk dataset & integrating RL with dialogue strategies to improve conversational diversity and robustness.

Built an offline edge-deployment pipeline using model compression and quantization, optimizing the conversational model for memory, latency, model size, and inference efficiency under resource constraints.

PROJECTS

AI/ML Developer, Sky-Scribe, FCRIT Vashi

[Sept '24 - April '25]

(Prof. Prachi Verma, FCRIT) | [Project Link](#)

Built a gesture-based air writing system using OpenCV and MediaPipe for real-time hand-pose tracking.

Integrated Gemini API for real-time equation recognition, solving 5+ complex expressions with 99% accuracy.

Delivered a contactless, hands-free tool for classrooms, applying vision-based human-AI interaction.

AI/ML Developer, Grammarly for Hindi, IIT Gandhinagar

[August '25 - December '25]

(Prof. Mayank Singh, IIT Gandhinagar) | [Project Link](#)

Fine-tuned Sarvam-1B,for Hindi Grammatical Error Correction & evaluated low-resource transfer on HiWiKiEdits.

Benchmarked against Transformer baselines using GLEU, Precision,Recall and $F_{0.5}$ to assess performance.

Expanded training with ~20K synthetic Hindi error pairs via direct noise injection and Round Trip Translation (RTT) noise and deployed a real-time Grammarly-style interface.

VMSim: A Configurable Virtual Memory & TLB Simulator, IIT Gandhinagar [Project Link](#)

[May'26 - July'26]

Designed and implemented a C++ virtual memory simulator modeling multi-level page-table lookup, address translation, and demand paging, supporting page sizes and physical frame counts.

Built a TLB caching layer and pluggable page-replacement engine (FIFO, LRU, clock, optimal) with a workload generator, enabling controlled experiments on page-fault rate vs memory size.

Empirically reproduced Belady's anomaly under FIFO and benchmarked TLB hit-rate and page-fault behavior across varying frame counts and access patterns, validating simulator correctness against theoretical OS models.

TECHNICAL SKILLS

Programming Languages: Python (Flask, Pandas, PyTorch), C, C++ (STL, Data Structures, Algorithms),SQL/MySQL.

Tools: Microsoft Power Apps, Git, LaTeX

Key Areas/Courses: Machine Learning, Deep Learning, Natural Language Processing, Operating Systems, Computer Networks, Data Structures & Algorithms.

ACHIEVEMENTS

GATE 2025 qualified with a score of 694 and All India Rank 1350.

Sky-Scribe Project selected for National Board of Accreditation Showcase and was implemented in classrooms for trials.

POSITIONS OF RESPONSIBILITY

Teaching Assistant | IIT Gandhinagar

[August '25 - Present]

Assisted in ES 242 (DSA), ES 199 (PO AI), CS 613 (NLP) by conducting labs and tutorials, mentoring students, grad-ing assignments, creating course assessments, mentoring in projects and supporting overall course coordination.

Technical Lead, AlgoZenith Club, FCRIT Vashi

[July '24 - May '25]

Led technical initiatives for AlgoZenith, the official coding club of FCRIT Vashi.